

Anomaly Rescue Mission Student Handout

TOK Cognitive Science Activity Bank • Natural Sciences • 30-40 min

Category	Natural Sciences	Time	30-40 min
Difficulty	Medium	ID	anomaly-rescue-mission

Big idea: Anomalies are not automatically errors; they are tests of method, theory, and judgement.

Student-Facing Prompt

Inspect Anomaly Rescue Mission Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Anomaly Rescue Mission Stimulus A	Student-facing stimulus 1: Plant growth mostly increases with light exposure, except one low-light plant grows unusually tall. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Anomaly Rescue Mission Stimulus B	Student-facing stimulus 2: Reaction time improves with practice, except one trial suddenly becomes much slower. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Anomaly Rescue Mission Reveal / Twist	Student-facing stimulus 3: Reaction time improves with practice, except one trial suddenly becomes much slower. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Anomalies are not automatically errors; they are tests of method, theory, and judgement.

Actual Lesson Questions

#	Question for students
1	After inspecting Anomaly Rescue Mission Stimulus A, what is your first knowledge claim?
2	Which exact detail from Anomaly Rescue Mission Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	When should anomalous evidence change a theory?

Ready-to-Use Examples

- Plant growth mostly increases with light exposure, except one low-light plant grows unusually tall.
- Reaction time improves with practice, except one trial suddenly becomes much slower.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Activity Steps

1. Give groups a simple model and a dataset that mostly fits it.
2. Students identify the anomalous result and list possible explanations.
3. Groups choose a response: discard, repeat, investigate, or revise the model.
4. Groups defend what evidence would justify their response.
5. Debrief how scientists should treat results that do not fit expectation.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: When should anomalous evidence change a theory?
- How can expectations help and hinder scientific judgement?
- What is the difference between ignoring an anomaly and responsibly investigating it?

Knowledge Questions

- When should anomalous evidence change a theory?
- How do expectations shape interpretation of data?
- What makes an outlier scientifically important?

Transfer Example

For example, run this activity with a surprising data point that students can explain, ignore, or use to revise theory. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...